

Kevin Mekulu

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RESEARCH PROFILE

Industrial and systems engineer developing quantitative methods to uncover the structure of human behavior and build interpretable measures of cognition and health. Research integrates stochastic and Markov modeling, nonlinear and recurrence methods, time-series analysis, statistical learning, and multimodal behavioral signals. Contributions span computational representations of speech and gaze, candidate biomarker discovery, and intelligent measurement systems. This program connects methodological development with prospective human studies, technology translation, and published conceptual frameworks for agentic AI in cognitive health.

EDUCATION

Expected May 2027 Ph.D. Candidate, Industrial Engineering, The Pennsylvania State University
Dissertation: Multisensing AI for Early Cognitive Health: Speech, Eye-Tracking, and Digital Biomarkers
Advisor: Hui Yang, Ph.D.
May 2018 M.S., Applied Computational Mathematics and Statistics, University of Notre Dame
May 2017 B.S., Applied Mathematics, cum laude, Virginia Tech; NCAA Division I athlete

RESEARCH EXPERIENCE AND LEADERSHIP

2024-Present Research Team Lead, Complex Systems Monitoring, Modeling and Control Laboratory, Penn State
• Supervise a multidisciplinary undergraduate research team spanning engineering, computer science, neuroscience, and health sciences.
2024-Present Founder & Chief Executive Officer, DementiAnalytics
• Translate speech- and language-based biomarker research into a 60-second cognitive screening and monitoring platform for senior-living environments.
• Lead an IRB-approved, prospective multi-site study with 45 residents, 180+ recordings, and five enrolled senior-living communities; preparing longitudinal reassessment to evaluate within-person biomarker change.
• Coordinate biomarker research with speech-language pathology, neuropsychology, senior-living, and engineering collaborators, including Nicole Etter, Ph.D.; Timothy Brearly, Ph.D.; and Stephanie Ruth Young, Ph.D., Northwestern University Feinberg School of Medicine.

RESEARCH INTERESTS

Computational modeling of cognition and behavior; stochastic and dynamical modeling of human systems; nonlinear, recurrence, and multifractal methods; time-series and multimodal behavioral analysis; interpretable statistical learning and AI; cognitive and mental health measurement; digital biomarkers; adaptive and agentic AI systems; prospective and longitudinal evaluation.

PEER-REVIEWED JOURNAL PUBLICATIONS

1. **Mekulu, K.**, Aqlan, F., & Yang, H. (2026). A 60-second interpretable voice model for early dementia screening. *PLOS Digital Health*, 5(7), e0001552 <https://doi.org/10.1371/journal.pdig.0001552>
2. **Mekulu, K.**, Aqlan, F., & Yang, H. (2026). Agentic neurotechnology: A multi-agent AI framework for next-generation dementia care. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*, 12, e70299 <https://doi.org/10.1002/trc2.70299>
3. **Mekulu, K.**, Aqlan, F., & Yang, H. (2026). A transparent four-feature speech model for depression screening applicable across clinical and community settings, including assisted-living environments. *Frontiers in Digital Health*, 7, 1675103 <https://doi.org/10.3389/fdgth.2025.1675103>
4. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). Character-level linguistic biomarkers for precision assessment of cognitive decline: A symbolic recurrence approach. *Frontiers in Aging Neuroscience*, 17, 1681124 <https://doi.org/10.3389/fnagi.2025.1681124>
5. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). CharMark: Character-level Markov modeling for interpretable linguistic biomarkers of cognitive decline. *Frontiers in Digital Health*, 7, 1659366 <https://doi.org/10.3389/fdgth.2025.1659366>
6. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). Agentic artificial intelligence in cognitive screening: A translational roadmap for dementia care. *Journal of Alzheimer's Disease Reports* <https://doi.org/10.1177/25424823251407989>
7. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). Reimagining dementia screening: A stakeholder-informed perspective on artificial intelligence, digital biomarkers, and real-world implementation. *Journal of Alzheimer's Disease Reports* <https://doi.org/10.1177/25424823251395310>
8. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). The mild cognitive impairment window for optimal Alzheimer's disease intervention. *Journal of Alzheimer's Disease Reports*, 9(1) <https://doi.org/10.1177/25424823251370768>

MANUSCRIPT UNDER REVIEW

9. **Mekulu, K.**, Aqlan, F., & Yang, H. From gaze transitions to digital biomarkers: Markov prediction of virtual-reality task performance. *IISE Transactions on Occupational Ergonomics and Human Factors* (Under review)

MANUSCRIPTS IN PREPARATION

10. **Mekulu, K.**, Etter, N., Brearly, T., & Yang, H. Speech timing measures are not interchangeable across elicitation methods in senior living residents. (Manuscript in preparation)
11. **Mekulu, K.**, Aqlan, F., & Yang, H. Nonlinear multifractality of gaze dynamics in immersive virtual reality. (Manuscript in preparation)

PREPRINTS

12. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). Automated detection of early-stage dementia using large language models: A comparative study on narrative speech. *medRxiv* <https://doi.org/10.1101/2025.06.06.25329081>
13. **Mekulu, K.** (2025). AI-enhanced memory list assessment system: Multi-dimensional analysis and automated strategy detection for next-generation cognitive screening. *medRxiv* <https://doi.org/10.1101/2025.10.02.25337192>
14. **Mekulu, K.** (2025). AI-enhanced animal naming as a digital biomarker for early cognitive screening. *medRxiv* <https://doi.org/10.1101/2025.11.10.25339942>

PEER-REVIEWED CONFERENCE PROCEEDINGS AND ABSTRACTS

15. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). Automated detection of early-stage dementia using large language models: A comparative study on narrative speech. *Alzheimer's & Dementia*, 21(S7), e108503 https://doi.org/10.1002/alz70861_108503
16. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). A novel AI-driven Markov chain approach to dementia detection. *Proceedings of the IISE Annual Conference*
17. **Mekulu, K.**, Aqlan, F., & Yang, H. (2024). AI and dementia diagnosis: Insights from linguistic patterns. *Proceedings of the IISE Annual Conference*
18. **Mekulu, K.**, Aqlan, F., & Yang, H. (2023). Intelligent detection of neurocognitive disorders based on physiological and speech signals in a virtual environment. *Proceedings of the IISE Annual Conference*
19. Iznurov, A., **Mekulu, K.**, Bello, K., et al. (2023). AI-driven virtual reality for mental health assessment. *Proceedings of the IISE Annual Conference*

EXTERNAL FUNDING AND GRANT ACTIVITY

- 2024 **Awarded: NSF National I-Corps Teams Award, \$50,000, IIP-2425827** - Entrepreneurial Lead and primary proposal writer
- 2026 **Submitted: NSF Translation to Practice - Translate Track (TTP-T), \$1,200,000** - Co-PI and primary proposal writer

PATENTS AND INTELLECTUAL PROPERTY

Five patent applications covering speech-, language-, and eye-tracking-based cognitive assessment technologies, including international application PCT/US2026/020992, filed March 26, 2026, with earliest priority dated April 7, 2025 (applicant: The Penn State Research Foundation).

1. **System and Method for Voice-Based Detection of Depression and Early Cognitive Impairment**
2. **System and Method for Analyzing Speech Patterns to Detect Neurodegenerative Cognitive Disorders**
3. **System and Method for Multi-Dimensional Analysis and Automated Strategy Detection for Next-Generation Cognitive Screening**
4. **System and Method for Learning Assessment via Visual Cognition**
5. **System and Method for Scoring Framework for Verbal Fluency Tasks in Early Cognitive Screening**

TEACHING EXPERIENCE

- Fall 2017 **Co-Instructor, ACMS 30440: Probability and Statistics**, University of Notre Dame
- Fall 2021 **Teaching Assistant, IE 453: Simulation Modeling for Decision Support**, Penn State
- Spring 2021 **Teaching Assistant, IE 330: Engineering Analytics**, Penn State

TEACHING INTERESTS

Probability and engineering statistics; stochastic modeling; simulation; time-series analysis; data analytics and machine learning; healthcare systems engineering; human factors and human-centered systems.

HONORS AND FELLOWSHIPS

2026 Leighton Riess Graduate Fellowship in Engineering, Penn State College of Engineering
2025 Forbes 30 Under 30 - Healthcare
2025 Fall 2025 IME Doctoral Student Award, \$10,000, Penn State Industrial Engineering
2025 Samsung NEXT Mobile Advance Finalist, selected from 500+ global applicants
2025 Penn State AI Society Award, Social Impact Category
2024-2025 Diefenderfer Fellowship for Researcher Entrepreneurs, Penn State College of Engineering (two-time recipient)
2024-2025 NIH NCATS Translational Training Program, Penn State Center for Medical Innovation
2024 Ben Franklin Technology Partners Startup Accelerator Finalist, \$1,000 award
2024 Microsoft for Startups Azure Credits, \$5,000
2023 Penn State Graduate Research Exhibition, 2nd Place, Engineering Category

MENTORSHIP AND LEADERSHIP

- Mentor 12+ undergraduate researchers working across engineering, computer science, neuroscience, and health sciences.
- Led 180+ stakeholder interviews through NSF I-Corps with clinicians, caregivers, senior-living operators, and technology stakeholders.

PROFESSIONAL SERVICE

- Reviewer, IEEE Journal of Biomedical and Health Informatics
- Reviewer, Frontiers in Dementia - Dementia Care
- Reviewer, Data & Policy

SELECTED PRESENTATIONS AND INVITED TALKS

2025 Alzheimer's Association International Conference (AAIC) - "Automated Detection of Early-Stage Dementia Using Large Language Models"
2025 IEEE BHI - "Character-Level Linguistic Biomarkers for Precision Assessment of Cognitive Decline"
2025 Penn State Center for Socially Responsible AI - "The Cure is Timing: How AI Voice Biomarkers Open the Treatment Window in Alzheimer's"
2025 IISE Annual Conference, Atlanta, GA - "A Novel AI-Driven Markov Chain Approach to Dementia Detection"
2024 IISE Annual Conference, Montreal, Canada - "AI and Dementia Diagnosis: Insights from Linguistic Patterns"

SELECTED MEDIA RECOGNITION AND PUBLIC IMPACT

2026 Penn State Research Q&A on AI, Cognition, and Dementia [\[Link\]](#)
2025 Forbes 30 Under 30 - Healthcare [\[Link\]](#)
2024 Penn State Profile on NSF I-Corps and Dementia Research Translation [\[Link\]](#)
2024 McKnight's Senior Living Coverage of Multimodal Cognitive-Screening Research [\[Link\]](#)